

DILLAN IMANS

Medical AI Researcher

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SUMMARY

Medical AI researcher and M.Sc. student in AI in Biomedicine at the Technical University of Munich, with first-author publications at EMA4MICCAI, IEEE BIBM, and Diagnostics. Works on data- and label-efficient methods, including knowledge distillation, unsupervised domain adaptation, and unpaired cross-modal learning, for diagnostic imaging and surgical vision.

EDUCATION

M.Sc. AI in Biomedicine, Technical University of Munich Imaging and Sensing focus	Oct 2026 – present
B.S. Computer Science & Engineering, Sungkyunkwan University 95.7 / 100 · Dean's List · Academic Excellence Scholarship	Aug 2022 - Jun 2026

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea - Built end-to-end retinal screening system (optic disc/cup segmentation & vessel analysis) and its backend; ONNX-optimized for low-latency inference, delivered to a clinical partner - Developed a two-phase unsupervised domain adaptation framework (frequency alignment + SAM-guided refinement) for cross-modality brain tumor segmentation; +9.0 Dice over strongest baseline across 6 BraTS 2021 settings - Built a graph-mediated distillation framework for unpaired MRI-to-fundus hypertension prediction across disjoint cohorts; +5.2 AUC, +3.2 AUPRC over best baseline, MRI-free at inference - Built a codebook-based distillation framework for unpaired cross-modal classification; +5.7 and +2.5 macro-F1 on OCT→fundus and CT→X-ray	Oct 2024 - Jun 2026
AI Researcher · Labren CUHK, Hong Kong - Built preprocessing infrastructure and annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips) - Curated a dataset and ran initial model experiments for a multimodal surgical robotics paper in collaboration with NVIDIA - Contributed to 3 manuscripts in surgical vision and multimodal robotics	Jun 2024 - Jun 2026
AI Researcher · Infolab SKKU, South Korea Explainable multi-layer dynamic ensemble for depression detection and severity assessment; 88.3% accuracy across 2,763 clinical records with SHAP interpretability, benchmarked against 20+ algorithms	Feb 2024 - Oct 2024

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

Peer-reviewed — First author

- Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction." *EMA4MICCAI 2026*.
- Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." *IEEE BIBM 2025*.
- Imans, D.**, Abuhmed, T., Alharbi, M., & El-Sappagh, S. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." *Diagnostics*, 14(21), 2385 (2024).

Under review — First author

- Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Shared Semantic Codebook Distillation for Unpaired Cross-Modal Medical Classification." *Under review, ACCV 2026*.

Co-authored

5. He, Y., Guo, P., Xu, M., Li, Z., **Imans, D.**, et al. "Cosmos-H-Surgical: Learning Surgical Robot Policies from Videos via World Modeling." *Preprint*.
6. Xu, M., Huang, Z., **Imans, D.**, Ye, Y., Zhang, X., & Dou, Q. "SAP-Bench: Benchmarking Multimodal Large Language Models in Surgical Action Planning." *Preprint*.
7. Xu, M., Shen, D., Zhang, J., Yip, H.C., Gao, Y., Chen, C., **Imans, D.**, et al. "Generalized Recognition of Basic Surgical Actions Enables Skill Assessment and Vision-Language-Model-based Surgical Planning." *Under review, npj Digital Surgery*.

SERVICE

Reviewer, EMA4MICCAI 2026 (MICCAI workshop)

PROJECTS

Vigilia Mundi — *vigiliamundi.com*

Live geospatial dashboard aggregating global disaster and epidemic data (GDACS, WHO) on a 12-hour ingest cycle. Built an idempotent upsert pipeline, a deck.gl/MapLibre visualization layer, and a source-adaptor architecture for adding new feeds. Next.js, FastAPI, PostgreSQL, GitHub Actions.

TECHNICAL SKILLS

Programming Languages: Python · C/C++ · Java · TypeScript

ML & Core: PyTorch · NumPy · Pandas · scikit-learn · OpenCV

Medical Imaging: MONAI · nnU-Net · SimpleITK · NiBabel · pydicom

Tooling & Infra: Git · GitHub Actions · Linux · Docker · Bash · HuggingFace · wandb · TensorBoard

Web: Next.js · Tailwind · FastAPI · PostgreSQL · Supabase

LANGUAGES

English, Indonesian (Fluent) · Mandarin, Korean (Conversational) · German (Beginner)